

Bikram Pandit

☎ 541-745-9562 • ✉ panditb@oregonstate.edu • 🌐 bikcrum.com
in bikcrum • 🔗 bikcrum • 💰 NXiLN3oAAAAJ

Professional Summary

PhD candidate specializing in multi-robot control, reinforcement learning, and sim-to-real transfer, with 5 publications at top-tier peer-reviewed robotics conferences (ICRA, CoRL), including an Oral and a Spotlight, a 1st-place finish in the DARPA TIAMAT Challenge, and policies deployed on 6+ real robot platforms spanning bipeds, quadrupeds, wheeled-bipeds, and humanoids.

Education

Oregon State University

PhD in Artificial Intelligence

Corvallis, Oregon

Sept 2021 – Jan 2027 (Expected)

Research focus: large-scale robot learning, decentralized control, sim-to-real transfer, and loco-manipulation systems

Rungta College of Engineering and Technology

Bachelor's in Computer Science and Engineering

India

Aug 2013 – June 2017

Awards and Honors

- **1st Place – DARPA TIAMAT Challenge** **2026**
Developed robust Spot policies for high-speed exploration, fall recovery on unpredictable terrain, and obstacle avoidance in cluttered settings; integrated with an autonomous planner for sequential task execution.
- **Best Paper Award Finalist (Cognitive Robotics)** **ICRA, 2024**
Learning Vision-Based Bipedal Locomotion for Challenging Terrain.
- **2nd Place – AgAID Institute Hackathon** **2023**
Competed against 32 teams from 8+ universities in a Microsoft-sponsored hackathon.

Research Experience

Dynamic Robotics and AI Lab, Oregon State University

Research Assistant

Corvallis, Oregon

Sept 2021 – Present

- Developed decentralized multi-robot control policies for cooperative payload transport and object manipulation, scaling across bipeds, quadruped-arm teams, and humanoids (CoRL 2024, ICRA 2026).
- Contributed to the design of vision-based RL policies for bipedal locomotion over challenging terrain using onboard camera and proprioception, without explicit state estimation (ICRA 2024, Best Paper Finalist).
- Contributed to masked-training strategies improving policy robustness to sensor dropout across locomotion and manipulation tasks (CoRL 2024).
- Co-developed a hyperparameter-free energy-minimization method for RL-based robot control, cutting energy use without sacrificing task performance (CoRL 2025, Oral).
- Built simulation and training infrastructure on IsaacLab, MjLab, and MuJoCo; contributed to open-source codebases including IsaacLab, RSL-RL, MuJoCo, and Weights & Biases.
- Deployed learned policies on 6+ real robot platforms, including Cassie, Digit, Spot, D1H wheeled-biped, Unitree Go2, and H1, with accelerated onboard inference for real-time control.

RAI Institute (formerly Boston Dynamics AI Institute)

Robotics Research Intern

Cambridge, Massachusetts

Sept 2024 – Jan 2025

- Developed multi-instance Isaac Sim infrastructure supporting heterogeneous robot articulations for scalable data generation and policy training across humanoid loco-manipulation research.
- Built unified imitation learning pipelines consolidating multiple specialized controllers into a single deployable policy.
- Validated natural walking and running behaviors in simulation and on real humanoid hardware.

Publications

- **Learning Multi-Humanoid Pickup and Transport via Decentralized Object-Centric Control** Under review
Object-centric control abstraction unifying single- and multi-robot humanoid pickup, transport, and handoff, validated on Digit V3 humanoids.
- **Multi-Quadruped Cooperative Object Transport: Learning Decentralized Pinch-Lift-Move** ICRA, 2026
Gripperless pinch-lift-move coordination for quadruped-arm teams of 2–10, validated on Unitree Go2 quadrupeds with D1 arms.
[Project](#) [Page](#) · [Code](#)
- **Non-conflicting Energy Minimization in RL Based Robot Control (Oral)** CoRL, 2025
Hyperparameter-free energy optimization for RL control with sim-to-real validation.
[Project](#) [Page](#) · [Code](#)
- **Learning Decentralized Multi-Biped Control for Payload Transport (Spotlight)** CoRL, 2024
Decentralized policy scaling rigid cooperative carrying to arbitrarily sized teams of bipeds sharing one carrier, validated on Cassie robots.
[Project](#) [Page](#) · [Code](#)
- **Learning Vision-Based Bipedal Locomotion for Challenging Terrain (Best Paper Finalist)** ICRA, 2024
Vision-conditioned sim-to-real locomotion robust to challenging terrain without explicit state estimation.
[arXiv](#)
- **Simple Masked Training Strategies Yield Control Policies That Are Robust to Sensor Failure** CoRL, 2024
Robust policy training improving resilience to sensor dropout in real-world deployments.
[Paper](#)

Patents

- **Systems and Methods for Decentralized Multi-Robot Cooperative Transport** Under review

Industry Experience

MiCROTEC Inc.

Machine Learning Intern

June 2023 – Aug 2023

- Accelerated large-scale vision model training by 80% through optimized data pipelines and system-level performance tuning.
- Developed robust segmentation models addressing class imbalance and deployment constraints for production systems.
- Integrated model training, evaluation, and deployment into automated ML DevOps pipelines using Docker, CI, and experiment tracking.

Digital Product Labs

Senior Software Developer

Dec 2017 – June 2021

- Designed and deployed production machine learning systems across mobile, web, and cloud platforms.
- Integrated machine learning components into consumer-facing products, improving performance and scalability in production.
- Partnered with product and engineering teams to scope and ship AI-driven features end to end.